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Safe Reinforcement Learning for Mixed-Autonomy Platoon Control

Jingyuan Zhou, Longhao Yan, Kaidi Yang

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***Abstract*— Deep reinforcement learning (DRL) has been demonstrated to be effective for the control of mixed-autonomy platoons. However, incorporating safety considerations into DRL typically relies on designing collision avoidance-based reward functions, which lack collision-free guarantees. To address this research gap, we introduce a differentiable safety layer to convert DRL actions to safe actions with collision-free guarantees, which relies on a differentiable quadratic programming problem featuring Control Barrier Function-based safety constraints. The proposed safety-informed DRL approach explicitly integrates safety constraints into the training process and enables our method to adapt to varying safety-critical scenarios. Simulation results demonstrate that our proposed method effectively ensures CAV safety in mixed platoon environments while simultaneously enhancing traffic capacity and string stability. Furthermore, the safety layer expedites training by reducing the range of exploration.**

I. Introduction

Connected and automated vehicles (CAVs) have been widely recognized as beneficial for traffic flow, thanks to their capability of exchanging real-time information and eliminating undesirable human behavior [1]–[3]. Significant research efforts have been placed on longitudinal control, i.e., the coordinated control of the speed profile of a platoon of CAVs, which has the potential to enhance road capacity [4], traffic stability [5], energy efficiency [6], and safety [3]. Early research on longitudinal control tends to assume all vehicles to be CAVs [7]. However, such an assumption is not practical in the near future since the penetration rates of CAVs can only increase gradually. Therefore, it is crucial to study mixed-autonomy platoons whereby CAVs and human-driven vehicles (HDVs) coexist.

In the context of mixed-autonomy platoons, several control schemes have been proposed to enable CAVs to consider the behavior and information of surrounding HDVs. One typical scheme is connected cruise control [8], whereby a CAV adjusts its driving decisions by considering multiple HDVs ahead. However, such a scheme only considers the preceding vehicles of the studied CAV without accounting for the impact on the following vehicles. In contrast, the recent-proposed scheme of leading cruise control (LCC) extends traditional cruise control by allowing CAVs to utilize information from both preceding and following HDVs [5]. With the additional information and flexibility compared to connected cruise control, LCC offers a better potential to improve the performance of the entire platoon.

Various controllers have been developed to implement LCC, including linear feedback controllers [5], optimization-based controllers based on model predictive control (MPC) and data-enabled predictive control (DeePC) [9], and learning-based controllers such as deep reinforcement learning (DRL) [10], [11]. Linear controllers are easy to design but can hardly handle complex constraints and optimization objectives. MPC can explicitly incorporate complex constraints and objectives into an embedded optimization problem, which, however, relies on explicit system modeling and requires accurate system identification to estimate system parameters. Although DeePC can simultaneously perform system identification and control via a data-driven non-parametric approach, it requires large amounts of computational resources to update the decision variables in each step, which limits its application. Among all these controllers, DRL-based controllers seem promising in that they can handle complex scenarios with a marginal online computational burden. However, existing DRL-based mixed-autonomy platoon control can only indirectly consider safety by incorporating safety-related penalties into the reward function, which can hardly provide safety guarantees.

Integrating safety guarantees into DRL-based control is an active research area with various algorithms being developed, such as reachability analysis-based approaches [12], safe exploration algorithms [13], and control barrier functions (CBF)-based approaches [14]–[17]. We focus on CBF-based approaches because CBF can be constructed as constraints on the control input and become active only when the safety condition is violated without significantly influencing the effectiveness of the controller. Recently, Reference [15] developed a CBF-quadratic programming (QP) problem that can compensate for the DRL action such that the compensated action is within a safe set. However, in this work, the CBF-QP is only considered as part of the environment without providing guidance on the training of the DRL model. In contrast, Reference [17] integrated a CBF-QP-based differentiable safety layer into neural network-based controllers, which can ensure safety while enabling back-propagation of the safety layer and thus facilitating the training of neural networks. However, this method was only applied in supervised learning settings to imitate an optimal controller, whereby the optimal control decisions are pre-calculated and used as labels. In other words, the method has not been adapted to model-free RL-based control strategies. Nevertheless, optimal control trajectories are often not easy to collect, especially when the surrounding vehicles' models are unknown. Moreover, these existing works have not been applied to safe RL for mixed-autonomy platoons. In this paper, we will devise a safety-informed DRL-based CAV control framework in the LCC context, whereby CBF is integrated into DRL via a differentiable quadratic program (QP) layer to encode safety guarantee into the training of DRL. We evaluate our proposed method with both Deep Deterministic Policy Gradient (DDPG) [18] and Proximal

The authors are with the Department of Civil and Environmental Engineering, National University of Singapore, Singapore 119077. Email:{jingyuanzhou, longhao.yan}@u.nus.edu, kaidi.yang@nus.edu.sg.

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Policy Optimization (PPO) [19] algorithms.

Statement of contribution. The contributions of this paper are two-fold. First, we design a safety-informed RL control strategy for CAVs in mixed-autonomy platoons by integrating CBF into DRL via a differentiable neural network layer. The proposed method provides safety guarantees for DRL while enhancing traffic efficiency. Furthermore, the incorporation of the differentiable safety layer expedites the training process, allowing for more rapid development and deployment of effective DRL-based CAV control strategies in real-world traffic situations. Second, we evaluate our proposed method with both DDPG and PPO in case studies with typical mixed-autonomy platoons, showcasing the effectiveness of our proposed method.

The rest of this paper is organized as follows. Section II introduces preliminaries about RL and CBFs. Section III presents the system modeling for mixed-autonomy platoon control. Section IV presents our methodological framework for LCC that integrates safety guarantees into DRL via a CBF-QP-based approach. Section V performs case studies to evaluate our method. Section VI concludes the paper.

II. Preliminaries

In this section, we introduce the concept and theoretical foundations underlying this paper in the context of Reinforcement Learning (RL) (Section II-A) and Control Barrier Function (CBF) (Section II-B).

A. Reinforcement Learning (RL)

In the RL context, a dynamic environment is modeled as a Markov decision process (MDP) $\mathcal{M} = (\mathcal{X}, \mathcal{U}, P, r, \gamma)$, where $\mathcal{X}$ and $\mathcal{U}$ denote the set of states and the set of actions, respectively, $P : \mathcal{X} \times \mathcal{U} \to \mathcal{X}$ represents the system dynamics in the form of a conditional probability $P(\boldsymbol{x}_{t+1}|\boldsymbol{x}_t, \boldsymbol{u}_t)$ with $\boldsymbol{x}_t, \boldsymbol{x}_{t+1} \in \mathcal{X}$ and $\boldsymbol{u}_t \in \mathcal{U}$, $r : \mathcal{X} \times \mathcal{U} \to \mathbb{R}$ represents the reward function, and $\gamma \in (0, 1]$ refers to a discount factor. The goal of RL is to learn a policy $\pi : \mathcal{X} \to \mathcal{U}$ in the form of a conditional probability $\pi(\boldsymbol{u}_t|\boldsymbol{x}_t)$ with $\boldsymbol{x}_t \in \mathcal{X}$ and $\boldsymbol{u}_t \in \mathcal{U}$ to maximize the discounted reward under this policy. Specifically, the RL objective function can be written $J(\pi) = \mathbb{E}_{\tau \sim p_\pi}\left[\sum_{t=0}^{T} \gamma^t r(\boldsymbol{x}_t, \boldsymbol{u}_t)\right]$, where $\tau = (\boldsymbol{x}_0, \boldsymbol{u}_0, \cdots, \boldsymbol{x}_T, \boldsymbol{u}_T)$ represents a trajectory defined as the sequence of states and actions of length T, and p_π denotes the distribution of trajectories under policy π. With the definition of MDP, we then introduce the concept of two RL algorithms DDPG and PPO, which are representatives of deterministic and stochastic DRL policies, respectively.

Both DDPG and PPO are constructed in an actor-critic structure, with an actor network π_θ producing an action based on the observed state, and a critic network estimating the state-action value and generating loss values during training.

In DDPG, the critic network Q_η gets updated based on the temporal difference (TD) error between the estimation and true value of the state-action pair:

$$\delta = r + \gamma Q_{\eta'}\left(x', \pi_{\theta'}\left(u' \mid x'\right)\right) - Q_\eta\left(x, u\right) \tag{1}$$

where γ denotes the discount factor, x' referrs to the next time step state, u' refers to the action at the next time step, θ' represents actor parameters at the next time step.

The actor network of DDPG is trained by updating its parameters using stochastic gradient ascent to maximize the expected return, with the gradient represented by

$$\nabla_\theta J\left(\pi_\theta\right) = \mathbb{E}_x\left[\nabla_\theta Q_\eta\left(x, \pi_\theta\left(x\right)\right)\right] \tag{2}$$

In PPO, the critic estimates the state value of CAV instead of the state-action value. Note that state-value can be converted to state-action value probabilistically following the Bellman equation [20]. The critic network, $V_\phi(x)$, estimates the value function of MDP and updates its parameters with the aim of minimizing the error between the predicted value function and the actual return:

$$\mathcal{L}(\phi) = \mathbb{E}_{x, a \sim \pi_\theta}\left[\delta^2\right], \quad \delta = r + \gamma V_\phi\left(x'\right) - V_\phi(x) \tag{3}$$

The actor network of PPO then gets updated following the loss function evaluated by the critic network:

$$L^{\text{clip}}(\pi_\theta) = \mathbb{E}_{x, u \sim \pi_{\theta_{\text{old}}}}\left[\min\left(\frac{\pi_\theta(u \mid x)}{\pi_{\theta_{\text{old}}}\left(u \mid x\right)} A^{\pi_{\theta_{\text{old}}}}\left(x, u\right), \text{clip}\left(\frac{\pi_\theta(u \mid x)}{\pi_{\theta_{\text{old}}}\left(u \mid x\right)}, 1-\epsilon, 1+\epsilon\right) A^{\pi_{\theta_{\text{old}}}\left(x, u\right)}\right)\right] \tag{4}$$

The function clip$(\cdot)$ prevents aggressive updating by utilizing a trust region characterized by ϵ. $A^{\pi_{\theta_{\text{old}}}\left(x, u\right)}$ is the advantage function evaluating the benefit of selected actions.

B. Control Barrier Function (CBF)

We next introduce CBF that incorporates safety conditions into constraints for control input. Specifically, consider a control affine system:

$$\dot{x} = f(x) + g(x)u, \tag{5}$$

where $\boldsymbol{x} \in \mathbb{R}^n$, $f : \mathbb{R}^n \to \mathbb{R}^n$ and $g : \mathbb{R}^n \to \mathbb{R}^{n \times m}$ are locally Lipschitz continuous, $u \in \mathbb{R}^m$.

Definition 1 (Control Barrier Function [21]): Let $\mathcal{C} \subset D \subset \mathbb{R}^n$, with $\mathcal{C} = \{x \in D \subset \mathbb{R}^n : h(x) \geq 0\}$ be the superlevel set of a continuously differentiable function $h : D \to \mathbb{R}$, then h is a control barrier function for system (5) if there exists an extended class $\mathcal{K}_\infty$ functions α_{CBF} such that:

$$\sup_{u \in U}\left[L_f h(x) + L_g h(x) u\right] \geq -\alpha_{\text{CBF}}(h(x)), \forall x \in D \tag{6}$$

CBFs (6) are constructed to ensure system safety by designing a forward-invariant safe set $\mathcal{C}$, which implies that if the current state resides within the safe set and the control input is constrained by the CBF, then the subsequent states are guaranteed to remain within the safe set.

III. Mixed-Autonomy Platoon Modeling

As illustrated in Fig. 1, we consider a mixed-autonomy platoon comprising HDVs (denoted by set $\Omega_{\mathcal{H}}$) and CAVs (denoted by set $\Omega_{\mathcal{C}}$), whereby the driving behavior of HDVs is modeled using car-following models, and CAVs are controlled in a distributed manner based on RL-based controllers. CAVs collect state information within a certain range from both preceding and following vehicles in sets $\Omega_{\mathcal{P},i}$ and $\Omega_{\mathcal{F},i}$ for CAV i, respectively, and determine their control actions based on the collected information. For each vehicle $i \in \Omega_{\mathcal{C}} \cup \Omega_{\mathcal{H}}$, we consider dynamics described by second-order ordinary differentiable equations with state

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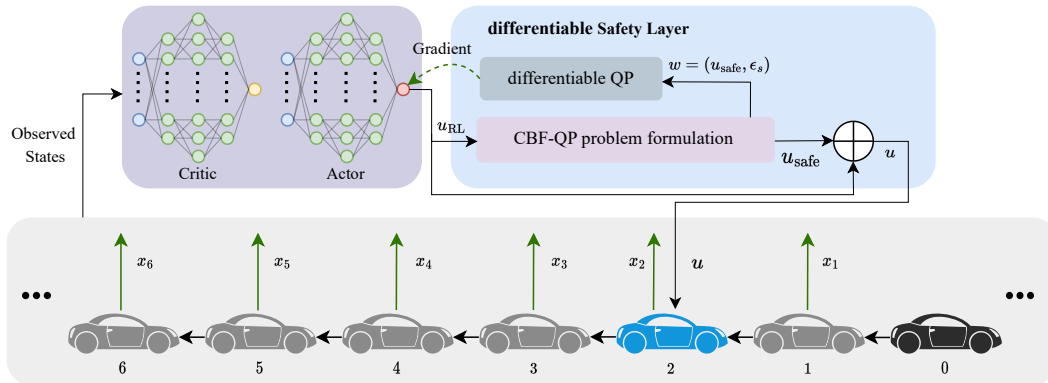

Fig. 1: Overview of the proposed controller for CAVs in mixed-autonomy traffic framework, the dotted line represents backpropagation of the differentiable QP.

including the speed of vehicle i (i.e., v_i) and the spacing between vehicle i and vehicle $i-1$ (i.e., s_i).

The dynamics of CAVs can be described as:

$$\begin{cases} \dot{s}_i(t) = v_{i-1}(t) - v_i(t), \\ \dot{v}_i(t) = u_i(t). \end{cases} \quad i \in \Omega_{\mathcal{C}}, \tag{7}$$

where the acceleration rate is the control input $u_i(t)$.

The dynamics of HDVs can be described as:

$$\begin{cases} \dot{s}_i(t) = v_{i-1}(t) - v_i(t), \\ \dot{v}_i(t) = \mathbb{F}\left(s_i(t), v_i(t), v_{i-1}(t)\right), \end{cases} \quad i \in \Omega_{\mathcal{H}}, \tag{8}$$

where the acceleration rate is determined by a car-following model $\mathbb{F}(\cdot)$ as a function of the spacing $s_i(t)$, the velocity of the preceding vehicle $v_{i-1}(t)$, and its own velocity $v_i(t)$.

Without loss of generality, we choose the optimal velocity model (OVM) [22] as the car-following model $\mathbb{F}$ following the setting of LCC [5]. The OVM can be replaced by any car-following model without influencing the applicability of the proposed method. The OVM reads as follows:

$$\mathbb{F}(\cdot) = \alpha\left(V\left(s_i(t)\right) - v_i(t)\right) + \beta(v_{i-1}(t) - v_i(t)), \tag{9}$$

where $\alpha, \beta > 0$ are positive constants representing car-following gains. $V(s)$ denotes the spacing-dependent desired velocity of HDVs with the form given in (10), where s_{st} and s_{go} represent the spacing thresholds for stopped and free-flow states, respectively, and $v_{\max}$ denotes the free-flow velocity.

$$V(s) = \begin{cases} 0, & s \leq s_{\text{st}} \\ \dfrac{v_{\max}}{2}\left(1 - \cos\left(\pi \dfrac{s - s_{\text{st}}}{s_{\text{go}} - s_{\text{st}}}\right)\right), & s_{\text{st}} < s < s_{\text{go}} \\ v_{\max}, & s \geq s_{\text{go}} \end{cases} \tag{10}$$

Combing the dynamics of CAVs (7) and HDVs (8), the longitudinal dynamics of mixed-traffic can be written as:

$$\dot{x}(t) = f(x(t), v_0(t)) + Bu(t), \tag{11}$$

where $x(t) = [s_1(t), v_1(t), \ldots, s_n(t), v_n(t)]^\top$ denotes the system states of all vehicles, $v_0(t)$ denotes the velocity of the head vehicle, and $f(\cdot)$ indicates vehicle dynamics including the nonlinear dynamics of HDVs and the linear dynamics of CAVs. System matrix B for CAVs' control input is summarized as $B = \left[e_{2n}^{2i_1}, e_{2n}^{2i_2}, \ldots, e_{2n}^{2i_m}\right] \in \mathbb{R}^{2n \times m}$, where n and m represent the number of vehicles and the number of CAVs, respectively, and $e_{2n}^{i} \in \mathbb{R}^{2n}$ associated with CAV i is a vector with the $2i$-th entry being 1 and the others 0.

IV. Safety-Critical Learning-based Control for CAV Platoons

As illustrated in Fig. 1, the method involves three modules: an RL module, a CBF-QP module, and a differentiable QP layer that facilitates the backpropagation of the QP module, where the CBF-QP module and differentiable QP layer form the differentiable safety layer. For each CAV (e.g., the blue car in Fig. 1), the RL module takes its observed states as input and gives an output of the control decision u_{RL}. Although the RL reward function includes safety considerations, there is no guarantee that u_{RL} is safe due to the black-box nature of RL. The CBF-QP module aims to address this issue by converting u_{RL} into a safe action. Specifically, a residual term u_{safe} is calculated by solving a CBF-QP to compensate u_{RL} such that the final action $u = u_{\text{RL}} + u_{\text{safe}}$ is a safe action. However, despite the CBF-QP module being able to provide a safety guarantee, it cannot guide the training of RL, i.e., providing gradient information into the backpropagation process. To address this issue, a differentiable QP layer is further added to calculate the gradients of the obtained QP solution (i.e., u_{safe}) with respect to u_{RL} by performing local sensitivity analysis on the KKT conditions of the QP. With such gradients, the DRL can improve its performance under safety constraints and expedite exploration time.

We next present the details of the three modules. Section IV-A designs the RL-based controller, Section IV-B incorporates safety guarantees into DRL via a CBF-QP module, and Section IV-C describes the differentiable QP layer that enables backpropagation of the CBF-QP module.

A. RL-based Controller Design

We formulate the control of mixed-autonomy platoons as an MDP $\mathcal{M} = (\mathcal{X}, \mathcal{U}, P, r, \gamma)$, where $\mathcal{X}$ includes all possible states $x(t)$ (i.e., location and velocity), $\mathcal{U}$ comprises all possible actions $u(t)$ (i.e., the acceleration rate), the process is defined as $P = (f, B)$ that involves the vehicle dynamics and control matrix. The reward function r is introduced next.

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Our control objective of each CAV is to stabilize traffic flow, improve traffic efficiency, and ensure its own safety. Therefore, the reward function of CAV i contains three parts:

$$R_i = r_{\text{stability},i} + r_{\text{efficiency},i} + r_{\text{safety},i} \tag{12}$$

where R_i represents the total reward, and $r_{\text{stability},i}$, $r_{\text{efficiency},i}$, and $r_{\text{safety},i}$ denote the stability reward, efficiency reward, and safety reward for CAV i, respectively. Next, these three types of rewards will be discussed in detail.

For stability, we focus on string stability [23], a desirable property of vehicle platoons such that disturbances of the lead vehicle do not get amplified when propagating through the platoon. Specifically, we aim to minimize velocity oscillations of both the CAV and the following vehicles resulting from the disturbances of the leading vehicle $i-1$:

$$r_{\text{stability},i} = -(v_i - v_{i-1})^2 - \sum_{j \in \Omega_{\mathcal{F},i}} \kappa_j (v_j - v_{i-1})^2 \tag{13}$$

where $\kappa_j \leq 1$ is the coefficient for the following vehicle velocity oscillation.

We characterize traffic efficiency as follows [24]:

$$r_{\text{efficiency},i} = \begin{cases} -1 & \text{TH}_i \geq 2.5 \\ 0 & \text{otherwise} \end{cases} \tag{14}$$

where $TH_i(t) = \frac{s_i(t)}{v_i(t)}$ approximates the headway. The goal is to increase flow represented as the reciprocal of headway. The headway threshold is set to ensure high flow on the road section.

For safety, we follow the common practice in existing works (e.g., [25]) and design a reward related to the time-to-collision (TTC) metric.

$$r_{\text{safety},i} = \begin{cases} \log\left(\frac{\text{TTC}_i}{4}\right) & 0 \leq \text{TTC} \leq 4 \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

where TTC is defined as $\text{TTC}_i(t) = -\frac{s_i(t)}{v_{i-1}(t) - v_i(t)}$. We set the time-to-collision threshold at 4 seconds, below which indicates dangerous driving and results in a negative reward.

B. CBF-QP-based Approach with Safety Guarantees

In this subsection, we introduce the CBF-QP formulation based on a CBF-based safety constraint defined in (6). To avoid collisions, it is essential for each CAV to maintain sufficient headway from the proceeding vehicle, and hence we can write the CBF candidate as

$$h(x_i) = s_i - \tau v_i. \tag{16}$$

where τ is the desired time headway, and the safety condition for CAV i can be expressed $h(x_i) \geq 0$.

Using the CBF definition (6) and Lie derivatives of the CBF candidate, the CBF constraint is given by:

$$L_f h(x_i) + L_g h(x_i) u + \alpha_{\text{CBF}}(h(x_i)) \geq 0 \tag{17}$$

where the class $\mathcal{K}$ function $\alpha_{\text{CBF}}(\cdot)$ is set to a linear function with a coefficient $\gamma_{\text{CBF}} > 0$. We then obtain the following:

$$\frac{1}{\tau}(v_{i-1} - v_i + \gamma_{\text{CBF}}(s_i - \tau v_i)) \geq u \tag{18}$$

To better tailor the derived CBF constraint to the DRL settings, we rewrite (18) as follows:

$$v_{i-1} - v_i - \tau(u_{\text{RL}} + u_{\text{safe}}) + \gamma_{\text{CBF}}(s_i - \tau v_i) \geq -\epsilon_s \tag{19}$$

where $u = u_{\text{RL}} + u_{\text{safe}}$ with u_{RL} representing the action provided by the DRL algorithm and u_{safe} the safe compensation, and $\epsilon_s \geq 0$ signifies a relaxation term to ensure feasibility and reduce the impact of CBF on the reward. Also note that we divide the control input into u_{RL} and u_{safe} to highlight the compensation effect of the CBF-QP, whereby u_{safe} and its corresponding gradients diminish with the training of DRL.

Moreover, due to the actuator limitations, we have the acceleration constraints as follows:

$$a_{\max} \geq u_{\text{RL}} + u_{\text{safe}} \geq a_{\min}. \tag{20}$$

where $a_{\max}$ and $a_{\min}$ denote the maximum and minimum acceleration rates for CAVs, respectively.

Next, we design the CBF-QP controller for CAVs. The objective is to minimize the control input deviation u_{safe} and relaxation term σ under the above-mentioned constraints. To this end, we form the QP optimization problem yielding

$$\begin{aligned} \min_{u_{\text{safe}}, \epsilon_s} \quad & ||u_{\text{safe}}||^2 + k\epsilon_s^2, \\ \text{subject to} \quad & (19), (20) \end{aligned} \tag{21}$$

where $k > 0$ is a penalty coefficient.

C. Differentiable QP for Neural Networks

In this subsection, we present a differentiable QP to enable the backpropagation of the QP solution derived in Section IV-B to facilitate the training of RL.

The loss function is denoted by ℓ, and the optimal solution of the CBF-QP is $w^\star = (u^\star_{\text{safe}}, \epsilon^\star_s)$, so the derivative of the loss can be written as $\frac{\partial \ell}{\partial w^\star}$. To train RL, we are interested in calculating the following partial derivative:

$$\frac{\partial \ell}{\partial \theta} = \frac{\partial \ell}{\partial w^\star} \frac{\partial w^\star}{\partial u_{\text{RL}}} \frac{\partial u_{\text{RL}}}{\partial \theta} \tag{22}$$

Since $\frac{\partial \ell}{\partial w^\star}$ can be easily obtained from the form of the loss function, and $\frac{\partial u_{\text{RL}}}{\partial \theta}$ is defined by the architecture of the RL embedded neural networks, we only need to calculate $\frac{\partial w^\star}{\partial u_{\text{RL}}}$. Let us rewrite the aforementioned CBF-QP (21) in a more general form with decision variable $w = (u_{\text{safe}}, \epsilon_s)$:

$$\begin{aligned} \min_{w} \quad & \frac{1}{2} w^T Q w \\ \text{subject to} \quad & Gw \leq q(u_{\text{RL}}) \end{aligned} \tag{23}$$

where Q, G, q are the corresponding parameters of the QP problem. Note that only q is a function of u_{RL}. As in [17], the calculation of $\frac{\partial w^\star}{\partial u_{\text{RL}}}$ can be performed by differentiating the KKT conditions with equality, i.e.,

$$\begin{aligned} Qw^\star + G^T \lambda^\star &= 0 \\ D(\lambda^\star)(Gw^\star - q(u_{\text{RL}})) &= 0, \end{aligned} \tag{24}$$

where $w^\star$ and $\lambda^\star$ represent the optimal primal and dual variables, and $D(\lambda^\star)$ is a matrix with $\lambda^\star$ at the diagonal.

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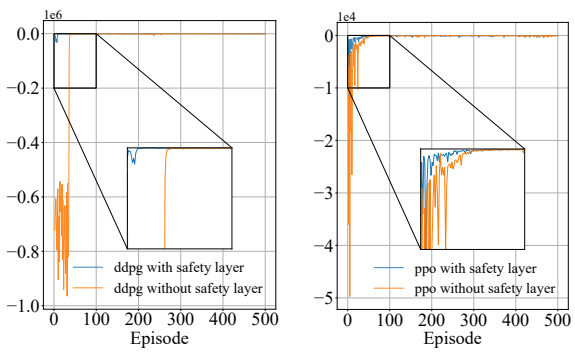
Then we take the derivative of the RL action u_{RL} and write in the matrix form as:

$$\begin{bmatrix} \dfrac{\partial w^\star}{\partial u_{\text{RL}}} \\ \dfrac{\partial \lambda^\star}{\partial u_{\text{RL}}} \end{bmatrix} = K^{-1} \begin{bmatrix} \mathbb{O} \\ D(\lambda^\star) \dfrac{\partial q(u_{\text{RL}})}{\partial u_{\text{RL}}} \end{bmatrix}. \tag{25}$$

with $K = \begin{bmatrix} Q & G^T \\ D(\lambda^\star) G & D(Gw^\star - q(u_{\text{RL}})) \end{bmatrix}$.

V. Simulation Results



(a) DDPG & safety-informed DDPG (b) PPO & safety-informed PPO

Fig. 2: Training rewards per episode.

In this section, we evaluate the developed method in simulations of a typical mixed-autonomy platoon. The simulation scenario is illustrated by Figure 1, which features a platoon of five vehicles traveling on a single lane, with the third vehicle being a CAV and the others HDVs.

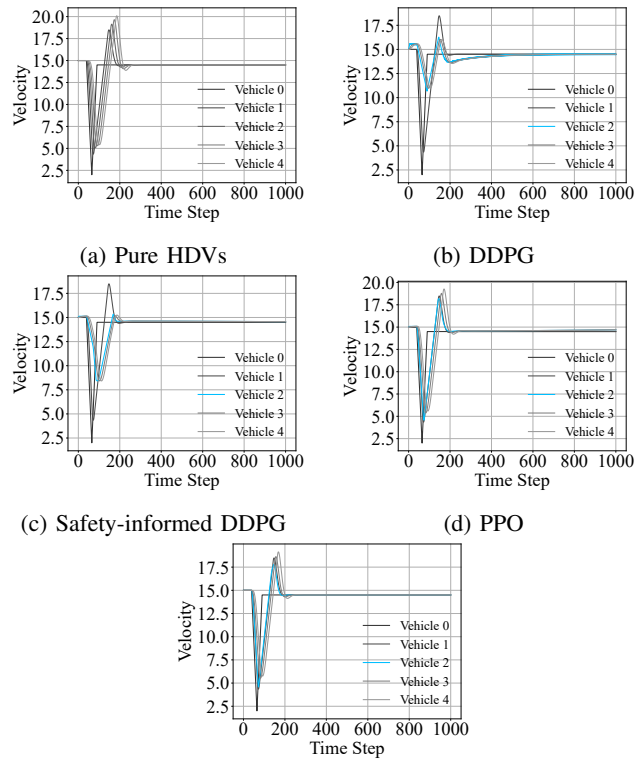
A. Training Settings and Results

In the training scenario, we utilize a random velocity disturbance setup, wherein the leading vehicle's velocity disturbance is sampled from a Gaussian distribution with zero mean and a standard deviation of 2 m/s independently at each time step within an episode. This chosen velocity disturbance is then multiplied by the time step (0.1s) and added to the head vehicle's original velocity. For the parameters for HDV car-following model (i.e., OVM), we consider $\alpha = 0.6, \beta = 0.9, s_{\text{st}} = 5, s_{\text{go}} = 35$. The equilibrium spacing and velocity are 20 m and 15 m/s.

For DDPG, the learning rates for the actor and critic are set at $1\text{e}{-4}$ and $1\text{e}{-3}$, respectively, while for PPO, both rates are $3\text{e}{-4}$, and we employ a linear decay schedule for the learning rate to facilitate training. As such, the performance of the algorithms is not sensitive to the initial learning rate and we choose the ones that can make the training process converge rapidly. The batch size for DDPG sampling is 8, and the reward decay is set to 0.99. For PPO, the batch size is 2048, the advantage function parameter is 0.95, the PPO clip parameter is set to 0.2, and the number of PPO epochs is set to 10. The training episodes are 500. For CBF parameters, time headway τ and γ_{CBF} are set to 0.3 s and 1, respectively.

Under the specified training scenarios and parameter settings, the training rewards per episode for DDPG, PPO, safety-informed DDPG, and safety-informed PPO are depicted in Figure 2. All algorithms converge after a sufficient number of training episodes, indicating that the designed reward function is suitable for mixed-autonomy platoon environments. Notably, the algorithms with "safety-informed" settings converge faster, which is expected as the safety layer reduces the range of feasible parameters and accelerates training, especially in scenarios with large parameter space.

B. Test Results



(a) Pure HDVs (b) DDPG

(c) Safety-informed DDPG (d) PPO

(e) Safety-informed PPO

Fig. 3: Vehicles' velocity for emergency braking scenario.

We examine an emergency braking scenario in which the leading vehicle initiates a sudden braking action with a deceleration of -5 m/s^2 for 2.5 seconds and then resumes its equilibrium velocity with the acceleration of 5 m/s^2 for the following 2.5 seconds. Note that this structured and extreme testing scenario is significantly different from the training scenarios with random disturbances. Such treatment is to demonstrate the performance of the safety layer under adversarial conditions and the generalizability of our model to unseen scenarios. We compare five types of models: pure HDVs, DDPG, PPO, safety-informed DDPG, and safety-informed PPO. Fig. 3 and 4 display the velocity and spacing at each time step in the test simulation. For both DDPG and PPO, the spacing tends to decrease to enhance traffic efficiency, and string stability improves as the velocity error is smaller compared to the pure HDV scenario. However, in Fig. 4 (b) and (d) without safety layers, safety is not guaranteed for DDPG and PPO, since the CAV under the DDPG algorithm collides with the leading vehicle, while the spacing for the PPO algorithm is also dangerously close. In

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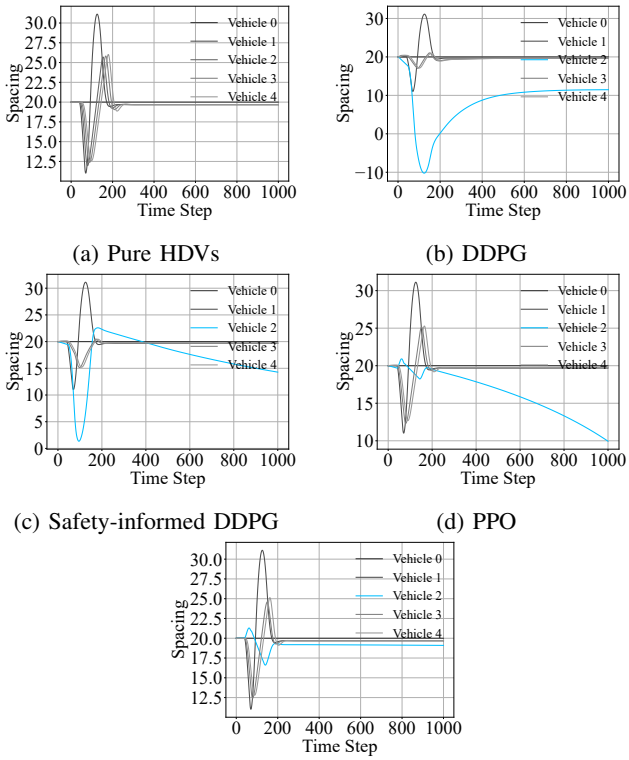

(a) Pure HDVs (b) DDPG

(c) Safety-informed DDPG (d) PPO

(e) Safety-informed PPO

Fig. 4: Vehicles' spacing for emergency braking scenario.

contrast, the safety-informed RL illustrated in Fig. 4 (c) and (e) is collision-free for the entire test process, making the driving strategies reliable and efficient.

VI. Conclusion

In this paper, we develop a safety-informed platoon controller by integrating DRL with a differentiable CBF layer, which has been shown by simulations to effectively enhance traffic efficiency and stability while providing safety guarantees. Furthermore, the safety layer contributes to faster exploration, which can be beneficial to the development and deployment of DRL algorithms. In future work, we will improve the generalizability in changing environments by integrating parameterized CBFs or meta-RL (e.g., [26]).

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